

5

Streaming instabilities and the Landau problem

5.1 Overview

The previous chapter demonstrated the existence of several types of plasma waves while Section 3.9 showed that particles can interchange energy with waves. The present chapter is devoted to establishing an important theory governing collisionless plasma waves, namely the theory of Landau damping. This theory characterizes the energy exchange that can occur between particles and waves when the wave-phase velocity is of the order of the particle thermal velocity. The Landau theory thus fills in the missing link between the adiabatic and isothermal susceptibilities discussed on p. 151. The first part of this chapter develops some preparatory ideas by showing that complex frequencies, i.e., wave instabilities, can develop if there are suitably arranged particle beams in a plasma. If one then considers a particle velocity distribution $f(v)$ as a collection of particle beams having different velocities, it should come as no surprise that waves with complex frequencies can occur. The next and main part of the chapter develops the detailed theory of Landau damping and summarizes the results in the form of a special function, called the “plasma dispersion function”. The last part of the chapter examines the Penrose criterion, a method for determining the stability properties of a given velocity distribution function $f(v)$.

5.2 Streaming instabilities

The electrostatic dispersion relation for a zero-temperature plasma is simply

$$1 - \sum_{\sigma} \frac{\omega_{p\sigma}^2}{\omega^2} = 0, \quad (5.1)$$

indicating that a *spatially-independent* oscillation at the plasma frequency

$$\omega_p = \sqrt{\omega_{pe}^2 + \omega_{pi}^2} \quad (5.2)$$

is a normal mode of a cold plasma. Once started, such an oscillation would persist indefinitely because no dissipative mechanism exists to quench it. On the other hand, the oscillation would have to be initiated by some source, because no available free energy exists from which the oscillation could draw on to start spontaneously.

We now consider a slightly different situation where cold electrons or ions stream at some spatially uniform initial velocity instead of being at rest in equilibrium. In the special situation where electrons and ions stream at the same velocity, the center of mass would also move at this velocity. One could then simply move to the center-of-mass frame where both species would be stationary and so, as argued in the previous paragraph, an oscillation would not start spontaneously. However, in the more general situation where the electrons and ions stream at *different* velocities, then both species have kinetic energy in the center-of-mass frame. This free energy could drive an instability.

In order to determine how such an instability could occur, the situation where each species has a specified equilibrium streaming velocity $\mathbf{u}_{\sigma 0}$ will now be examined. The linearized equation of motion, continuity equation, and Poisson's equation respectively become

$$\frac{\partial \mathbf{u}_{\sigma 1}}{\partial t} + \mathbf{u}_{\sigma 0} \cdot \nabla \mathbf{u}_{\sigma 1} = -\frac{q_{\sigma}}{m_{\sigma}} \nabla \phi_1, \quad (5.3)$$

$$\frac{\partial n_{\sigma 1}}{\partial t} + \mathbf{u}_{\sigma 0} \cdot \nabla n_{\sigma 1} = -n_{\sigma 0} \nabla \cdot \mathbf{u}_{\sigma 1}, \quad (5.4)$$

and

$$\nabla^2 \phi_1 = -\frac{1}{\epsilon_0} \sum_{\sigma} q_{\sigma} n_{\sigma 1}. \quad (5.5)$$

As before, all first-order dependent variables are assumed to vary as $\exp(i\mathbf{k} \cdot \mathbf{x} - i\omega t)$. Combining the equation of motion and the continuity equation gives

$$n_{\sigma 1} = n_{\sigma 0} \frac{k^2}{(\omega - \mathbf{k} \cdot \mathbf{u}_{\sigma 0})^2} \frac{q_{\sigma}}{m_{\sigma}} \phi_1. \quad (5.6)$$

Substituting this into Eq. (5.5) gives the dispersion relation

$$1 - \sum_{\sigma} \frac{\omega_{p\sigma}^2}{(\omega - \mathbf{k} \cdot \mathbf{u}_{\sigma 0})^2} = 0, \quad (5.7)$$

which is just like the susceptibility for stationary cold species except that here ω is replaced by the Doppler-shifted frequency $\omega_{\text{Doppler}} = \omega - \mathbf{k} \cdot \mathbf{u}_{\sigma 0}$.

Two examples of streaming instability will now be considered: (i) equal densities of positrons and electrons streaming past each other with equal and opposite velocities, and (ii) electrons streaming past stationary ions.

Positron–electron streaming instability

The positron/electron example, while difficult to realize in practice, is worth analyzing because it reveals essential features of the streaming instability with a minimum of mathematical effort. The equilibrium positron and electron densities are assumed equal so as to have charge neutrality. Since electrons and positrons have identical mass, the positron plasma frequency ω_{pp} is the same as the electron plasma frequency ω_{pe} . Let $\mathbf{u}_0$ be the electron stream velocity and $-\mathbf{u}_0$ be the positron stream velocity. Defining $z = \omega/\omega_{pe}$ and $\lambda = \mathbf{k} \cdot \mathbf{u}_0/\omega_{pe}$, Eq. (5.7) reduces to

$$1 = \frac{1}{(z - \lambda)^2} + \frac{1}{(z + \lambda)^2}, \quad (5.8)$$

a quartic equation in z . Because of the symmetry, no odd powers of z appear and Eq. (5.8) becomes

$$z^4 - 2z^2(\lambda^2 + 1) + \lambda^4 - 2\lambda^2 = 0, \quad (5.9)$$

which may be solved for z^2 to give

$$z^2 = (\lambda^2 + 1) \pm \sqrt{4\lambda^2 + 1}. \quad (5.10)$$

Each choice of the $\pm$ sign gives two roots for z . If $z^2 > 0$, then the two roots are real, equal in magnitude, and opposite in sign. On the other hand, if $z^2 < 0$, then the two roots are *pure imaginary*, equal in magnitude, and opposite in sign. Recalling that $\omega = \omega_{pe}z$ and that the perturbation varies as $\exp(i\mathbf{k} \cdot \mathbf{x} - i\omega t)$, it is seen that the positive imaginary root corresponds to instability; i.e., to a perturbation that *grows* exponentially in time.

Hence the condition for instability is $z^2 < 0$. Because only the choice of minus sign in Eq. (5.10) allows this possibility, we select the minus sign and determine that the condition for instability is

$$\sqrt{4\lambda^2 + 1} > \lambda^2 + 1, \quad (5.11)$$

which corresponds to

$$0 < \lambda < \sqrt{2}. \quad (5.12)$$

The maximum growth rate is found by maximizing the right-hand side of Eq. (5.10) with the minus sign chosen. Taking the derivative with respect to λ and setting $dz/d\lambda = 0$ to find the maximum, gives

$$2z \frac{dz}{d\lambda} = 2\lambda - \frac{4\lambda}{\sqrt{4\lambda^2 + 1}} = 0 \quad (5.13)$$

or $\lambda = \sqrt{3}/2$. Substituting this most unstable λ back into Eq. (5.10) (with the minus sign selected, since this choice gives the potentially unstable root) gives the maximum growth rate to be $y = 1/2$, where $z = x + iy$.

Changing back to physical variables, it is seen that onset of instability occurs when

$$ku_0 < \sqrt{2}\omega_{pe}, \quad (5.14)$$

and the maximum growth rate occurs when

$$ku_0 = \frac{\sqrt{3}}{2}\omega_{pe}, \quad (5.15)$$

in which case

$$\omega = i\frac{\omega_{pe}}{2}. \quad (5.16)$$

Figure 5.1 plots the normalized instability growth rate $\text{Im } z$ as a function of λ ; both onset and maximum growth rate are indicated. Since the instability has a pure imaginary frequency it is called a purely growing mode. Because the growth rate is of the order of magnitude of the plasma frequency, the instability grows extremely quickly.

Electron-ion streaming instability

Now consider the more realistic situation where electrons stream with velocity $\mathbf{v}_0$ through a background of stationary neutralizing ions. The dispersion relation in this case is

$$1 - \frac{\omega_{pi}^2}{\omega^2} - \frac{\omega_{pe}^2}{(\omega - \mathbf{k} \cdot \mathbf{u}_0)^2} = 0, \quad (5.17)$$

which can be recast in non-dimensional form by defining $z = \omega/\omega_{pe}$, $\epsilon = m_e/m_i$, and $\lambda = \mathbf{k} \cdot \mathbf{u}_0/\omega_{pe}$, giving

$$1 = \frac{\epsilon}{z^2} + \frac{1}{(z - \lambda)^2}. \quad (5.18)$$

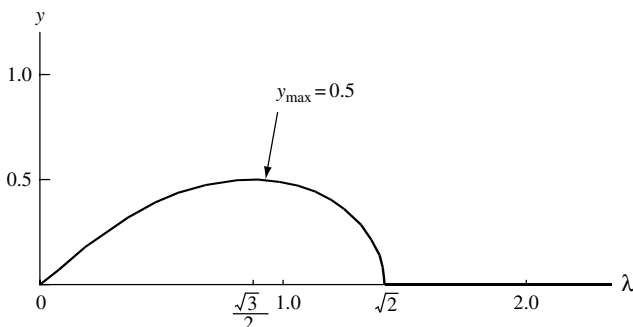


Fig. 5.1 Normalized growth rate vs. normalized wavenumber.

The value of λ at which onset of instability occurs is found by plotting the right-hand side (RHS) of Eq. (5.18) versus z and comparing this to the left-hand side, which always has the value of unity. This plot is shown in the sequence Fig. 5.2(a)–(c). The first term ϵ/z^2 diverges at $z = 0$, while the second term diverges at $z = \lambda$. Between $z = 0$ and $z = \lambda$, the right-hand side of Eq. (5.18) has a minimum. If the value of the right-hand side at this minimum is below unity, as in Fig. 5.2(a), there will be two places between $z = 0$ and $z = \lambda$ where the right-hand side of Eq. (5.18) equals unity. For $z > \lambda$, there is always one and only one place where the right-hand side equals unity and similarly for $z < 0$ there is one and only one place where the right-hand side equals unity. Thus, when the minimum of the right-hand side is below unity, Eq. (5.18) has four real roots. If λ is decreased to some critical value, then, as shown in Fig. 5.2(b), the two roots located between 0 and λ coalesce. If λ is decreased still more, then, as shown in Fig. 5.2(c), the minimum of the right-hand side is above unity so there are only two real roots (those in the regions $z > \lambda$ and $z < 0$). In this latter case the other two roots of this quartic equation must be complex.

Because a quartic equation must be expressible in the form

$$(z - z_1)(z - z_2)(z - z_3)(z - z_4) = 0, \quad (5.19)$$

and because the coefficients of Eq. (5.18) are real, the two complex roots must be complex conjugates of each other. To see this, suppose the complex roots are

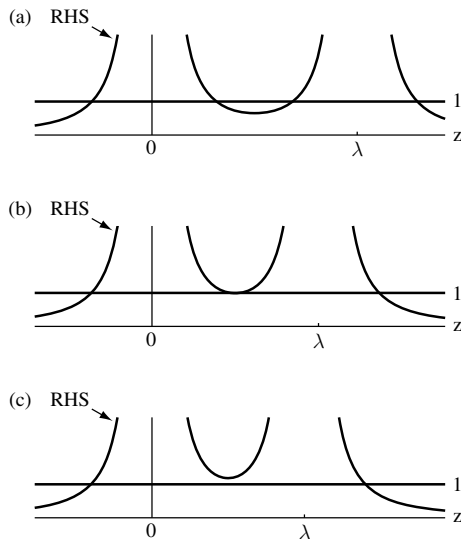


Fig. 5.2 Plot of right-hand side (RHS) and left-hand side (unity) of Eq. (5.18) for sequence of λ values. (a) Four real roots, (b) middle two roots coalesce, (c) two real roots only.

z_1 and z_2 and the real roots are z_3 and z_4 . The product of the first two factors in Eq. (5.19) is $z^2 - (z_1 + z_2)z + z_1z_2$; if the complex roots are not complex conjugates of each other then this product will contain complex coefficients and, when multiplied with the product of the terms involving the real roots, will result in an equation that contains complex coefficients. However, Eq. (5.18) has only real coefficients so the two complex roots must be complex conjugates of each other. The complex root with positive imaginary part will give rise to instability.

Thus, when the minimum of the right-hand side of Eq. (5.18) is greater than unity, two of the roots become complex, and one of these complex roots gives instability. The onset of instability occurs when the minimum of the right-hand side of Eq. (5.18) equals unity. Straightforward analysis (cf. assignments) shows this occurs when

$$\lambda = (1 + \epsilon^{1/3})^{3/2}, \quad (5.20)$$

i.e., instability starts when

$$\mathbf{k} \cdot \mathbf{u}_0 = \omega_{pe} \left[1 + \left(\frac{m_e}{m_i} \right)^{1/3} \right]^{3/2}. \quad (5.21)$$

The maximum growth rate of the instability may be found by solving Eq. (5.18) for λ and then simplifying the resulting expression using ϵ as a small parameter. The details of this are worked out in the assignments, where it is demonstrated that the maximum growth rate is

$$\max \omega_i \simeq \frac{\sqrt{3}}{2} \left(\frac{m_e}{2m_i} \right)^{1/3} \omega_{pe}, \quad (5.22)$$

which occurs when

$$\mathbf{k} \cdot \mathbf{u}_0 \simeq \omega_{pe}. \quad (5.23)$$

As before, this is a very fast-growing instability, about one order of magnitude smaller than the electron plasma frequency.

Streaming instabilities are a reason why certain simple proposed methods for attaining thermonuclear fusion will not work. These methods involve shooting an energetic deuterium beam at an oppositely directed energetic tritium beam with the expectation that collisions between the two beams would produce fusion reactions. However, such a system is extremely unstable with respect to the two-stream instability. This instability typically has a growth rate much faster than the fusion reaction rate and so will destroy the beams before significant fusion reactions can occur.

5.3 The Landau problem

A plasma wave behavior of great philosophical interest and of great practical importance can now be investigated. Before doing so, we recall three seemingly disconnected results obtained thus far, namely:

1. When the exchange of energy between charged particles and a simple one-dimensional electrostatic wave with dependence $\sim \exp(ikx - i\omega t)$ was considered, the particles were categorized into two general classes, trapped and untrapped, and it was found that untrapped particles tended to be dragged towards the wave phase velocity. Thus, untrapped particles moving slower than the wave gain kinetic energy, whereas untrapped particles moving faster lose kinetic energy. This has the consequence that if there are more slow than fast particles, the particles gain net kinetic energy overall and this gain presumably comes at the expense of the wave. Conversely, if there are more fast than slow particles, net energy flows from the particles to the wave.
2. When electrostatic plasma waves in an unmagnetized, uniform, stationary plasma were considered, it was found that wave behavior was characterized by a dispersion relation $1 + \chi_e(\omega, k) + \chi_i(\omega, k) = 0$, where $\chi_\sigma(\omega, k)$ is the susceptibility of each species σ . As sketched in Fig. 4.1 these susceptibilities had simple limiting forms when $\omega/k \ll \sqrt{\kappa T_{\sigma 0}/m_\sigma}$ (isothermal limit) and when $\omega/k \gg \sqrt{\kappa T_{\sigma 0}/m_\sigma}$ (adiabatic limit), but the fluid analysis failed when $\omega/k \sim \sqrt{\kappa T_{\sigma 0}/m_\sigma}$ and the susceptibilities became undefined.
3. When the behavior of interacting beams of particles was considered, it was found that under certain conditions a fast-growing instability would develop.

The analysis of the Landau problem, to be presented in the remainder of this chapter, will show that these three results are both interrelated and part of a larger picture.

5.3.1 Attempt to solve the linearized Vlasov–Poisson system of equations using Fourier analysis

The method for manipulating fluid equations to find wave solutions was as follows: (i) the relevant fluid equations were linearized, (ii) a perturbation $\sim \exp(i\mathbf{k} \cdot \mathbf{x} - i\omega t)$ was assumed, (iii) the system of partial differential equations was transformed into a system of algebraic equations, and then finally (iv) the roots of the determinant of the system of algebraic equations provided the dispersion relations that characterized the various wave solutions.

It seems reasonable to use this method again in order to investigate waves from the Vlasov point of view. However, it will be seen that this approach *fails* and that, instead, a more complicated Laplace transform technique must be used. However, once the underlying difference between the Laplace and Fourier transform techniques has been identified, it is possible to go back and “patch

up” the Fourier technique. Although perhaps not entirely elegant, this patching approach turns out to be a reasonable compromise in that it incorporates both the simplicity of the Fourier method and the correct mathematics/physics of the Laplace method.

The Fourier method will now be presented and, to highlight how this method fails, the simplest relevant example will be considered, namely a one-dimensional, unmagnetized plasma with a stationary Maxwellian equilibrium. The ions are assumed to be so massive as to be immobile and the ion density is assumed to equal the electron equilibrium density. The electrostatic electric field $E = -\partial\phi/\partial x$ is therefore zero in equilibrium because there is charge neutrality in equilibrium. Since ions do not move there is no need to track ion dynamics. Thus, all perturbed quantities refer to electrons and so it is redundant to label these with a subscript “e.” In order to have a well-defined, physically meaningful problem, the equilibrium electron velocity distribution is assumed to be Maxwellian, i.e.,

$$f_0(v) = n_0 \frac{1}{\pi^{1/2} v_T} e^{-v^2/v_T^2}, \quad (5.24)$$

where $v_T \equiv \sqrt{2\kappa T/m}$.

The one-dimensional, unmagnetized Vlasov equation is

$$\frac{\partial f}{\partial t} + v \frac{\partial f}{\partial x} - \frac{q}{m} \frac{\partial \phi}{\partial x} \frac{\partial f}{\partial v} = 0 \quad (5.25)$$

and linearization of this equation gives

$$\frac{\partial f_1}{\partial t} + v \frac{\partial f_1}{\partial x} - \frac{q}{m} \frac{\partial \phi_1}{\partial x} \frac{\partial f_0}{\partial v} = 0. \quad (5.26)$$

Because the Vlasov equation describes evolution in phase-space, v is an *independent* variable just like x and t . Assuming a normal mode dependence $\sim \exp(ikx - i\omega t)$, Eq. (5.26) becomes

$$-i(\omega - kv)f_1 - ik\phi_1 \frac{q}{m} \frac{\partial f_0}{\partial v} = 0, \quad (5.27)$$

which gives

$$f_1 = -\frac{k}{(\omega - kv)} \frac{q}{m} \frac{\partial f_0}{\partial v} \phi_1. \quad (5.28)$$

The electron density perturbation is

$$n_1 = \int_{-\infty}^{\infty} f_1 dv = -\frac{q}{m} \phi_1 \int_{-\infty}^{\infty} \frac{k}{(\omega - kv)} \frac{\partial f_0}{\partial v} dv, \quad (5.29)$$

a relationship between n_1 and ϕ_1 . Another relationship between n_1 and ϕ_1 is Poisson's equation

$$\frac{\partial^2 \phi_1}{\partial x^2} = -\frac{n_1 q}{\epsilon_0}. \quad (5.30)$$

Replacing $\partial/\partial x$ by ik , Eq. (5.30) becomes

$$k^2 \phi_1 = \frac{n_1 q}{\epsilon_0}. \quad (5.31)$$

Combining Eqs. (5.29) and (5.31) gives the dispersion relation

$$1 + \frac{q^2}{k^2 m \epsilon_0} \int_{-\infty}^{\infty} \frac{k}{(\omega - kv)} \frac{\partial f_0}{\partial v} dv = 0. \quad (5.32)$$

This can be written more elegantly by substituting for f_0 using Eq. (5.24), defining the non-dimensional particle velocity $\xi = v/v_T$, and the non-dimensional phase velocity $\alpha = \omega/kv_T$ to give

$$1 - \frac{1}{2k^2 \lambda_D^2} \frac{1}{\pi^{1/2}} \int_{-\infty}^{\infty} d\xi \frac{1}{(\xi - \alpha)} \frac{\partial}{\partial \xi} e^{-\xi^2} = 0 \quad (5.33)$$

or

$$1 + \chi = 0, \quad (5.34)$$

where the electron susceptibility is

$$\chi = -\frac{1}{2k^2 \lambda_D^2} \frac{1}{\pi^{1/2}} \int_{-\infty}^{\infty} d\xi \frac{1}{(\xi - \alpha)} \frac{\partial}{\partial \xi} e^{-\xi^2}. \quad (5.35)$$

In contrast to the earlier two-fluid wave analysis, where in effect the zeroth, first, and second moments of the Vlasov equation were combined (continuity equation, equation of motion, and equation of state), here only the Vlasov equation is involved. Thus the Vlasov equation contains all the information of the moment equations and more. The Vlasov method therefore seems a simpler and more direct way for calculating the susceptibilities than the fluid method, except for a serious difficulty: the integral in Eq. (5.35) is mathematically ill-defined because the denominator vanishes when $\xi = \alpha$ (i.e., when $\omega = kv_T$). Because it is not clear how to deal with this singularity, the ξ integral cannot be evaluated and the Fourier method fails. This is essentially the same as the problem encountered in fluid analysis when ω/k became comparable to $\sqrt{\kappa T/m}$.

5.3.2 Landau method: Laplace transforms

Landau (1946) argued that the Fourier problem presented above is ill-posed and showed that the linearized Vlasov–Poisson problem should be treated as an *initial-value* problem, rather than as a normal mode problem. The initial-value point of

view is conceptually related to the analysis of single particle motion in sawtooth or sine waves. Before presenting the Landau analysis of the linearized Vlasov–Poisson problem, certain important features of Laplace transforms will now be reviewed.

The Laplace transform of a function $\psi(t)$ is defined as

$$\tilde{\psi}(p) = \int_0^{\infty} \psi(t) e^{-pt} dt \quad (5.36)$$

and can be considered as a “half of a Fourier transform” since the time integration starts at $t = 0$ rather than $t = -\infty$. Caution is required regarding the convergence of this integral for situations where $\psi(t)$ contains exponentially growing terms.

Suppose such exponentially growing terms exist. As $t \rightarrow \infty$, the fastest growing term, say $\exp(\gamma t)$, will dominate all other terms contributing to $\psi(t)$. The integral in Eq. (5.36) will then diverge as $t \rightarrow \infty$, unless a *restriction* is imposed on the real part of p . In particular, if it is *required* that $\text{Re } p > \gamma$, then the decaying $\exp(-pt)$ factor will always overwhelm the growing $\exp(\gamma t)$ factor so that the integral in Eq. (5.36) will converge. These issues of convergence are ignored in Fourier transforms where it is implicitly assumed that the function being transformed has neither exponentially growing terms (which diverge at $t = \infty$) nor exponentially decaying terms (which diverge at $t = -\infty$).

Thus, the integral transform in Eq. (5.36) is defined *only* for $\text{Re } p > \gamma$. To emphasize this restriction, Eq. (5.36) is rewritten as

$$\tilde{\psi}(p) = \int_0^{\infty} \psi(t) e^{-pt} dt, \quad \text{Re } p > \gamma, \quad (5.37)$$

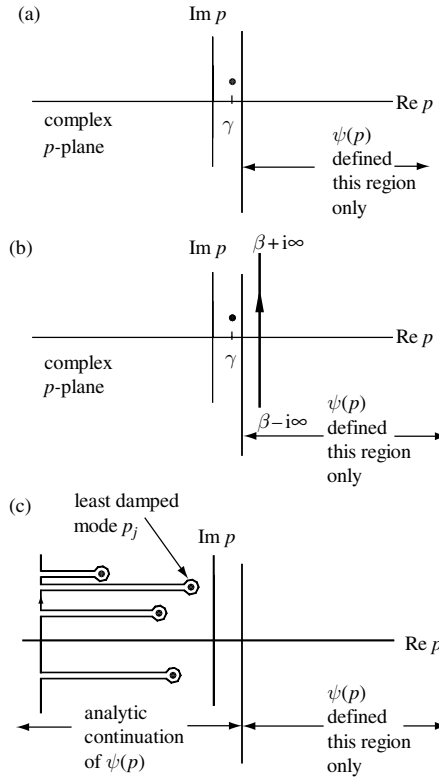
where γ is the fastest growing exponential term contained in $\psi(t)$. Since p is typically complex, Eq. (5.37) means that $\tilde{\psi}(p)$ is *only* defined in that part of the complex p -plane lying to the *right* of γ as sketched in Fig. 5.3(a). Whenever $\tilde{\psi}(p)$ is used, one must be very careful to avoid venturing outside the region in p -space where $\tilde{\psi}(p)$ is defined (this restriction will later become an important issue).

To construct an inverse transform, consider the integral

$$g(t) = \int_C dp \tilde{\psi}(p) e^{pt}. \quad (5.38)$$

This integral is ambiguously defined for now because the integration contour C is unspecified. However, whatever integration contour is ultimately selected must not venture into regions where $\tilde{\psi}(p)$ is undefined. Thus, an allowed integration path must have $\text{Re } p > \gamma$. Substitution of Eq. (5.37) into Eq. (5.38) and interchanging the order of integration gives

$$g(t) = \int_0^{\infty} dt' \int_C dp \psi(t') e^{p(t-t')}, \quad \text{Re } p > \gamma. \quad (5.39)$$

Fig. 5.3 Contours in complex p -plane.

A useful integration path C for the p integral will now be determined. Recall from the theory of Fourier transforms that the Dirac delta function can be expressed as

$$\delta(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} d\omega e^{i\omega t}, \quad (5.40)$$

which is an integral along the real ω axis so that ω is always real. The integration path for Eq. (5.39) will now be chosen such that the real part of p stays constant, say at a value β that is larger than γ , while the imaginary part of p goes from $-\infty$ to ∞ . This path is shown in Fig. 5.3(b), and is called the Bromwich contour.

For this choice of path, Eq. (5.39) becomes

$$\begin{aligned} g(t) &= \int_0^\infty dt' \int_{\beta-i\infty}^{\beta+i\infty} d(p_r + ip_i) \psi(t') e^{(p_r + ip_i)(t-t')} \\ &= i \int_0^\infty dt' e^{\beta(t-t')} \psi(t') \int_{-\infty}^\infty dp_i e^{ip_i(t-t')} \end{aligned}$$

$$\begin{aligned}
&= 2\pi i \int_0^\infty dt' e^{\beta(t-t')} \psi(t') \delta(t-t') \\
&= 2\pi i \psi(t),
\end{aligned} \tag{5.41}$$

where Eq. (5.40) has been used. Thus, $\psi(t) = (2\pi i)^{-1} g(t)$ and so the inverse of the Laplace transform is

$$\psi(t) = \frac{1}{2\pi i} \int_{\beta-i\infty}^{\beta+i\infty} dp \psi(p) e^{pt}, \quad \beta > \gamma. \tag{5.42}$$

Before returning to physics, recall another peculiarity of Laplace transforms, namely the transformation procedure for derivatives. The Laplace transform of $d\psi/dt$ may be simplified by integrating by parts to give

$$\int_0^\infty dt \frac{d\psi}{dt} e^{-pt} = [\psi(t) e^{-pt}]_0^\infty + p \int_0^\infty dt \psi(t) e^{-pt} = p\tilde{\psi}(p) - \psi(0). \tag{5.43}$$

Unlike Fourier transforms, here the *initial value* forms part of the transform. Thus, Laplace transforms contain information about the initial value and so should be better suited than Fourier transforms for investigating initial value problems. The importance of the initial value was also evident in the Chapter 3 analysis of particle motion in sawtooth or sine wave potentials.

The requisite mathematical tools are now in hand for investigating the Vlasov–Poisson system and its dependence on initial value. To obtain extra insights with little additional effort, the analysis is extended to the more general situation of a three-dimensional plasma where ions are allowed to move. Again, electrostatic waves are considered, and it is assumed that the equilibrium plasma is stationary, spatially uniform, neutral, and unmagnetized.

The equilibrium velocity distribution of each species is assumed to be a three-dimensional Maxwellian distribution function

$$f_{\sigma 0}(\mathbf{v}) = n_\sigma \left(\frac{m_\sigma}{2\pi\kappa T_\sigma} \right)^{3/2} \exp(-m_\sigma v^2/2\kappa T_\sigma). \tag{5.44}$$

The equilibrium electric field is assumed to be zero so that the equilibrium potential is a constant chosen to be zero. It is further assumed that at $t = 0$ there exists a small perturbation of the distribution function and that this perturbation evolves in time so that at later times

$$f_\sigma(\mathbf{x}, \mathbf{v}, t) = f_{\sigma 0}(\mathbf{v}) + f_{\sigma 1}(\mathbf{x}, \mathbf{v}, t). \tag{5.45}$$

The linearized Vlasov equation for each species is therefore

$$\frac{\partial f_{\sigma 1}}{\partial t} + \mathbf{v} \cdot \nabla f_{\sigma 1} - \frac{q_\sigma}{m_\sigma} \nabla \phi_1 \cdot \frac{\partial f_{\sigma 0}}{\partial \mathbf{v}} = 0. \tag{5.46}$$

All perturbed quantities are assumed to have the spatial dependence $\sim \exp(\mathbf{ik} \cdot \mathbf{x})$; this is equivalent to Fourier transforming in space. Equation (5.46) becomes

$$\frac{\partial f_{\sigma 1}}{\partial t} + \mathbf{ik} \cdot \mathbf{v} f_{\sigma 1} - \frac{q_{\sigma}}{m_{\sigma}} \phi_1 \mathbf{ik} \cdot \frac{\partial f_{\sigma 0}}{\partial \mathbf{v}} = 0. \quad (5.47)$$

Laplace transforming in time gives

$$(p + \mathbf{ik} \cdot \mathbf{v}) \tilde{f}_{\sigma 1}(\mathbf{v}, p) - f_{\sigma 1}(\mathbf{v}, 0) - \frac{q_{\sigma}}{m_{\sigma}} \tilde{\phi}_1(p) \mathbf{ik} \cdot \frac{\partial f_{\sigma 0}}{\partial \mathbf{v}} = 0, \quad (5.48)$$

which may be solved for $\tilde{f}_{\sigma 1}(\mathbf{v}, p)$ to give

$$\tilde{f}_{\sigma 1}(\mathbf{v}, p) = \frac{1}{(p + \mathbf{ik} \cdot \mathbf{v})} \left[f_{\sigma 1}(\mathbf{v}, 0) + \frac{q_{\sigma}}{m_{\sigma}} \tilde{\phi}_1(p) \mathbf{ik} \cdot \frac{\partial f_{\sigma 0}}{\partial \mathbf{v}} \right]. \quad (5.49)$$

This is similar to Eq. (5.28), except that now the Laplace variable p occurs instead of the Fourier variable $-\mathrm{i}\omega$ and also the initial value $f_{\sigma 1}(\mathbf{v}, 0)$ appears. As before, Poisson's equation can be written as

$$\nabla^2 \phi_1 = -\frac{1}{\epsilon_0} \sum_{\sigma} q_{\sigma} n_{\sigma 1} = -\frac{1}{\epsilon_0} \sum_{\sigma} q_{\sigma} \int d^3 v f_{\sigma 1}(\mathbf{x}, \mathbf{v}, t). \quad (5.50)$$

Replacing $\nabla \rightarrow \mathbf{ik}$ and Laplace transforming with respect to time, Poisson's equation becomes

$$k^2 \tilde{\phi}_1(p) = \frac{1}{\epsilon_0} \sum_{\sigma} q_{\sigma} \int d^3 v \tilde{f}_{\sigma 1}(\mathbf{v}, p). \quad (5.51)$$

Substitution of Eq. (5.49) into the right-hand side of Eq. (5.51) gives

$$k^2 \tilde{\phi}_1(p) = \frac{1}{\epsilon_0} \sum_{\sigma} q_{\sigma} \int d^3 v \left\{ \frac{f_{\sigma 1}(\mathbf{v}, 0) + \frac{q_{\sigma}}{m_{\sigma}} \tilde{\phi}_1(p) \mathbf{ik} \cdot \frac{\partial f_{\sigma 0}}{\partial \mathbf{v}}}{(p + \mathbf{ik} \cdot \mathbf{v})} \right\}, \quad (5.52)$$

which is similar to Eq. (5.32) except that $-\mathrm{i}\omega \rightarrow p$ and the initial value appears. Equation (5.52) may be solved for $\tilde{\phi}_1(p)$ to give

$$\tilde{\phi}_1(p) = \frac{N(p)}{D(p)}, \quad (5.53)$$

where the numerator is

$$N(p) = \frac{1}{k^2 \epsilon_0} \sum_{\sigma} q_{\sigma} \int d^3 v \frac{f_{\sigma 1}(\mathbf{v}, 0)}{(p + \mathbf{ik} \cdot \mathbf{v})} \quad (5.54)$$

and the denominator is

$$D(p) = 1 - \frac{1}{k^2} \sum_{\sigma} \frac{q_{\sigma}^2}{\epsilon_0 m_{\sigma}} \int d^3 v \frac{\mathbf{ik} \cdot \frac{\partial f_{\sigma 0}}{\partial \mathbf{v}}}{(p + \mathbf{ik} \cdot \mathbf{v})}. \quad (5.55)$$

Note that the denominator is similar to Eq. (5.32). All that has to be done now is take the inverse Laplace transform of Eq. (5.53) to obtain

$$\phi_1(t) = \frac{1}{2\pi i} \int_{\beta-i\infty}^{\beta+i\infty} dp \frac{N(p)}{D(p)} e^{pt}, \quad (5.56)$$

where β is chosen to be larger than the fastest growing exponential term in $N(p)/D(p)$.

This is an exact formal solution to the problem. However, because of the complexity of $N(p)$ and $D(p)$ it is impossible to evaluate the integral in Eq. (5.56). Nevertheless, it turns out to be feasible to evaluate the long-time asymptotic limit of this integral and, for practical purposes, this is a sufficient answer to the problem.

5.3.3 The relationship between poles, exponential functions, and analytic continuation

Before evaluating Eq. (5.56), it is useful to examine the relationship between exponentially growing/decaying functions, Laplace transforms, poles, residues, and analytic continuation. This relationship is demonstrated by considering the exponential function

$$f(t) = e^{qt}, \quad (5.57)$$

where q is a complex constant. If the real part of q is positive, then the amplitude of $f(t)$ is exponentially growing, whereas if the real part of q is negative, the amplitude of $f(t)$ is exponentially decaying. Now, calculate the Laplace transform of $f(t)$; it is

$$\tilde{f}(p) = \int_0^\infty e^{(q-p)t} dt = \frac{1}{p-q}, \quad \text{defined only for } \operatorname{Re} p > \operatorname{Re} q. \quad (5.58)$$

Let us examine the Bromwich contour integral for $\tilde{f}(p)$ and temporarily call this integral $F(t)$; evaluation of $F(t)$ ought to yield $F(t) = f(t)$. Thus, we define

$$F(t) = \frac{1}{2\pi i} \int_{\beta-i\infty}^{\beta+i\infty} dp \tilde{f}(p) e^{pt}, \quad \beta > \operatorname{Re} q. \quad (5.59)$$

If the Bromwich contour could be closed in the left-hand p -plane, the integral could easily be evaluated using the method of residues but closure of the contour to the left is forbidden because of the restriction that $\beta > \operatorname{Re} q$. This annoyance may be overcome by constructing a new function $\hat{f}(p)$ that:

1. equals $\tilde{f}(p)$ in the region $\beta > \operatorname{Re} q$,
2. is *also* defined in the region $\beta < \operatorname{Re} q$, and
3. is analytic.

Integration of $\hat{f}(p)$ along the Bromwich contour gives the same result as does integration of $\tilde{f}(p)$ along the same contour because the two functions are *identical* along this contour (cf. stipulation (1) above). Thus, it is seen that

$$F(t) = \frac{1}{2\pi i} \int_{\beta-i\infty}^{\beta+i\infty} dp \hat{f}(p) e^{pt}, \quad (5.60)$$

but now there is no restriction on which part of the p -plane may be used. So long as the end points are kept fixed and no poles are crossed, the path of integration of an analytic function can be arbitrarily deformed. This is because the difference between the original path and a deformed path is a closed contour, which integrates to zero if it does not enclose any poles. Because $\hat{f}(p) \rightarrow 0$ at the endpoints $\beta \pm \infty$, the integration path of $\hat{f}(p)$ can be deformed into the left-hand plane as long as $\hat{f}(p)$ remains analytic (i.e., does not jump over any poles or branch cuts). How can this magic function $\hat{f}(p)$ be constructed?

The answer is simple: we *define* a function $\hat{f}(p)$ having the identical functional form as $\tilde{f}(p)$, but *without* the restriction that $\text{Re } p > \text{Re } q$. Thus, the analytic continuation of

$$\tilde{f}(p) = \frac{1}{p-q}, \quad \text{defined only for } \text{Re } p > \text{Re } q, \quad (5.61)$$

is simply

$$\hat{f}(p) = \frac{1}{p-q}, \quad \text{defined for all } p, \text{ provided } \hat{f}(p) \text{ remains analytic.} \quad (5.62)$$

The Bromwich contour can now be deformed into the left-hand plane as shown in Fig. 5.4. Because $\exp(pt) \rightarrow 0$ for positive t and negative $\text{Re } p$, the integration contour can be closed by an arc that goes to the left (cf. Fig. 5.4) into the region where $\text{Re } p \rightarrow -\infty$. The resulting contour encircles the pole at $p = q$ and so the integral can be evaluated using the method of residues as follows:

$$F(t) = \frac{1}{2\pi i} \oint \frac{1}{p-q} e^{pt} dp = \lim_{p \rightarrow q} 2\pi i (p-q) \left[\frac{1}{2\pi i (p-q)} e^{pt} \right] = e^{qt}. \quad (5.63)$$

This simple example shows that while the Bromwich contour formally gives the inverse Laplace transform of $\tilde{f}(p)$, the Bromwich contour by itself does not allow use of the method of residues, since the poles of interest are located in the left-hand complex p -plane where $\tilde{f}(p)$ is undefined. However, analytic continuation of $f(p)$ allows deformation of the Bromwich contour into the formerly forbidden area, and then the inverse transform may be easily evaluated using the method of residues.

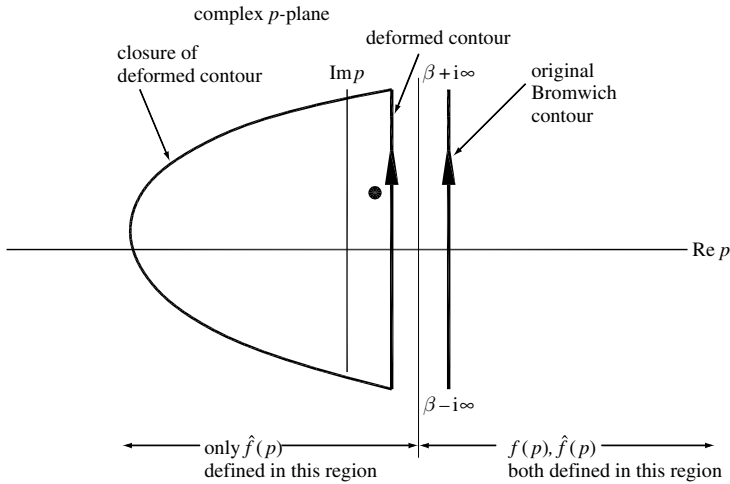


Fig. 5.4 Bromwich contour.

5.3.4 Asymptotic long-time behavior of the potential oscillation

We now return to the more daunting problem of evaluating Eq. (5.56). As in the simple example above, the goal is to close the contour to the left but, because the functions $N(p)$ and $D(p)$ are not defined for $\text{Re } p < \gamma$, this is not immediately possible. It is first necessary to construct analytic continuations of $N(p)$ and $D(p)$ that extend the definition of these functions into regions of negative $\text{Re } p$. As in the simple example, the desired analytic continuations may be constructed by taking the same formal expressions as obtained before, but now extending the definition to the entire p -plane with the proviso that *the functions remain analytic* as the region of definition is pushed leftwards in the p -plane.

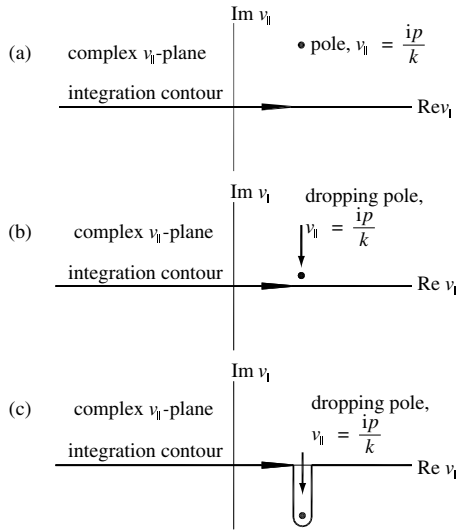
Consider first construction of an analytic continuation for the function $N(p)$. This function can be written as

$$N(p) = \frac{1}{k^2 \varepsilon_0} \sum_{\sigma} q_{\sigma} \int_{-\infty}^{\infty} dv_{\parallel} \frac{F_{\sigma 1}(v_{\parallel}, 0)}{(p + ikv_{\parallel})} = \frac{1}{ik^3 \varepsilon_0} \sum_{\sigma} q_{\sigma} \int_{-\infty}^{\infty} dv_{\parallel} \frac{F_{\sigma 1}(v_{\parallel}, 0)}{(v_{\parallel} - ip/k)}. \quad (5.64)$$

Here, $\parallel$ means in the $\mathbf{k}$ direction, and the parallel component of the initial value of the perturbed distribution function has been defined as

$$F_{\sigma 1}(v_{\parallel}, 0) = \int d^2 \mathbf{v}_{\perp} f_{\sigma 1}(\mathbf{v}, 0). \quad (5.65)$$

The integrand in Eq. (5.54) has a pole at $v_{\parallel} = ip/k$. Let us assume that $k > 0$ (the coordinate system can always be defined so that this is so). Before we construct an analytic continuation, $\text{Re } p$ is restricted to be greater than γ so that the pole $v_{\parallel} = ip/k$ is in the *upper half* of the complex $v_{\parallel}$ -plane as shown in Fig. 5.5(a).

Fig. 5.5 Complex $v_{\parallel}$ -plane.

When $N(p)$ is analytically continued to the left-hand region, the definition of $N(p)$ is extended to allow $\text{Re } p$ to become less than γ and even negative. As shown in Fig. 5.5(b), decreasing $\text{Re } p$ means that the pole at $v_{\parallel} = ip/k$ in Eq. (5.54) drops from its initial location in the upper half $v_{\parallel}$ -plane towards the lower half $v_{\parallel}$ -plane. A critical question now arises: how should we arrange this construction when $\text{Re } p$ passes through zero? If the pole is allowed to jump from being above the path of $v_{\parallel}$ integration (which is along the real $v_{\parallel}$ axis) to being below, the function $N(p)$ will *not* be analytic because it will have a discontinuous jump of $2\pi i$ times the residue associated with the pole. Since it was stipulated that $N(p)$ must be analytic, the pole cannot be allowed to jump over the $v_{\parallel}$ contour of integration. Instead, the prescription proposed by Landau will be used, which is to *deform* the $v_{\parallel}$ contour as $\text{Re } p$ becomes negative so that the contour *always* lies below the pole; this deformation is shown in Fig. 5.5(c).

$D(p)$ involves a similar integration along the real $v_{\parallel}$ axis. It also has a pole that is initially in the upper half-plane when $\text{Re } p > 0$, but then drops to being below the axis as $\text{Re } p$ is allowed to become negative. Thus, analytic continuation of $D(p)$ is also constructed by deforming the path of the $v_{\parallel}$ integration so that the contour always lies below the pole.

Equipped with these suitably constructed analytic continuations of $N(p)$ and $D(p)$ into the left-hand p -plane, evaluation of Eq. (5.56) can now be undertaken. As shown in the simple example, it is computationally advantageous to deform the Bromwich contour into the left-hand p -plane. The deformed contour evaluates

to the same result as the original Bromwich contour (provided the deformation does not jump over any poles) and this evaluation may be accomplished via the method of residues. In the general case where $N(p)/D(p)$ has several poles in the left-hand p -plane, then, as shown in Fig. 5.3(c), the contour may be deformed so that the vertical portion is pushed to the far left, except where there is a pole p_j ; the contour “snags” around each pole p_j as shown in Fig. 5.3(c). For $\text{Re } p \rightarrow -\infty$, the numerator $N(p) \rightarrow 0$, while the denominator $D(p) \rightarrow 1$. Since $\exp(pt) \rightarrow 0$ for $\text{Re } p \rightarrow -\infty$ and positive t , the left-hand vertical line does not contribute to the integral and Eq. (5.56) simply consists of the sum of the residues of all the poles, i.e.,

$$\phi_1(t) = \sum_j \lim_{p \rightarrow p_j} \left[(p - p_j) \frac{N(p)}{D(p)} e^{pt} \right]. \quad (5.66)$$

Where do the poles p_j come from? Upon examining Eq. (5.66), it is clear that poles could come either from (i) $N(p)$ having an explicit pole, i.e., $N(p)$ contains a term $\sim 1/(p - p_j)$, or (ii) from $D(p)$ containing a factor $\sim (p - p_j)$, i.e., p_j is a root of the equation $D(p) = 0$. The integrand in Eq. (5.64) has a pole in the $v_{\parallel}$ -plane; this pole is “used up” as a residue upon performing the $v_{\parallel}$ integration, and so does not contribute a pole to $N(p)$. The only other possibility is that the initial value $F_{\sigma 1}(v_{\parallel}, 0)$ somehow provides a pole, but $F_{\sigma 1}(v_{\parallel}, 0)$ is a physical quantity with a bounded integral i.e., $\int F_{\sigma 1}(v_{\parallel}, 0) dv_{\parallel}$ is finite and so cannot contribute a pole in $N(p)$. It is therefore concluded that all poles in $N(p)/D(p)$ must come from the roots (also called zeros) of $D(p)$.

The problem can be simplified by deciding to be content with a *less* than complete solution. Instead of attempting to calculate $\phi_1(t)$ for all positive times (i.e., all the poles p_j contribute to the solution), we restrict ourselves to the less burdensome problem of finding the long-time asymptotic behavior of $\phi_1(t)$. Because each term in Eq. (5.66) has a factor $\exp(ip_j t)$, the least damped term (i.e., the term with pole furthest to the right in Fig. 5.3(c)) will dominate all the other terms at large t . Hence, in order to find the long-time asymptotic behavior, all that is required is to find the root p_j having the largest real part.

The problem is thus reduced to finding the roots of $D(p)$; this requires performing the $v_{\parallel}$ integration sketched in Fig. 5.5(c). Before doing this, it is convenient to integrate out the perpendicular velocity dependence from $D(p)$ so that

$$\begin{aligned} D(p) &= 1 - \frac{1}{k^2} \sum_{\sigma} \frac{q_{\sigma}^2}{\epsilon_0 m_{\sigma}} \int d^3 v \frac{\mathbf{i} \mathbf{k} \cdot \frac{\partial f_{\sigma 0}}{\partial \mathbf{v}}}{(p + \mathbf{i} \mathbf{k} \cdot \mathbf{v})} \\ &= 1 - \frac{1}{k^2} \sum_{\sigma} \frac{q_{\sigma}^2}{\epsilon_0 m_{\sigma}} \int_{-\infty}^{\infty} dv_{\parallel} \frac{\frac{\partial F_{\sigma 0}}{\partial v_{\parallel}}}{(v_{\parallel} - \mathbf{i} p/k)}. \end{aligned} \quad (5.67)$$

Thus, the relation $D(p) = 0$ can be written in terms of susceptibilities as

$$D(p) = 1 + \chi_i + \chi_e = 0, \quad (5.68)$$

since the quantities being summed in Eq. (5.67) are essentially the electron and ion perturbations associated with the oscillation, and $D(p)$ is the Laplace transform analog of the Fourier transform of Poisson's equation. In the special case where the equilibrium distribution function is Maxwellian, the susceptibilities can be written in a standardized form as

$$\begin{aligned} \chi_\sigma &= -\frac{1}{2k^2\lambda_{D\sigma}^2} \frac{1}{\pi^{1/2}} \int_{-\infty}^{\infty} d\xi \frac{1}{(\xi - ip/kv_{T\sigma})} \frac{\partial}{\partial \xi} \exp(-\xi^2) \\ &= \frac{1}{k^2\lambda_{D\sigma}^2} \left[\frac{1}{\pi^{1/2}} \int_{-\infty}^{\infty} d\xi \frac{(\xi - ip/kv_{T\sigma} + ip/kv_{T\sigma})}{(\xi - ip/kv_{T\sigma})} \exp(-\xi^2) \right] \\ &= \frac{1}{k^2\lambda_{D\sigma}^2} \left[1 + \frac{1}{\pi^{1/2}} \alpha \int_{-\infty}^{\infty} d\xi \frac{\exp(-\xi^2)}{(\xi - \alpha)} \right] \\ &= \frac{1}{k^2\lambda_{D\sigma}^2} [1 + \alpha Z(\alpha)], \end{aligned} \quad (5.69)$$

where $\alpha = ip/kv_{T\sigma}$, and the last line introduces the *plasma dispersion* function $Z(\alpha)$ defined as

$$Z(\alpha) \equiv \frac{1}{\pi^{1/2}} \int_{-\infty}^{\infty} d\xi \frac{\exp(-\xi^2)}{(\xi - \alpha)}, \quad (5.70)$$

where the ξ integration path is under the dropped pole.

5.3.5 Evaluation of the plasma dispersion function

If the pole corresponding to the fastest growing (i.e., least damped) mode turns out to have dropped well below the real axis (corresponding to $\text{Re } p$ being large and negative), the fastest growing mode would be highly damped. We argue that this does not happen because there ought to be a correspondence between the Vlasov and fluid models in regimes where both are valid. Since the fluid model indicated the existence of undamped plasma waves when ω/k was much larger than the thermal velocity, the Vlasov model should predict nearly the same wave in this regime. The fluid wave model had no damping and so any damping introduced by the Vlasov model should be weak in order to maintain an approximate correspondence between fluid and Vlasov models. The Vlasov solution corresponding to the fluid mode can therefore have a pole only slightly below the real axis, i.e., only slightly negative. In this case, it is only necessary to analytically continue the definition of $N(p)/D(p)$ *slightly* into the negative p -plane. Thus, the pole in Eq. (5.70) drops only slightly below the real axis as shown in Fig. 5.6.

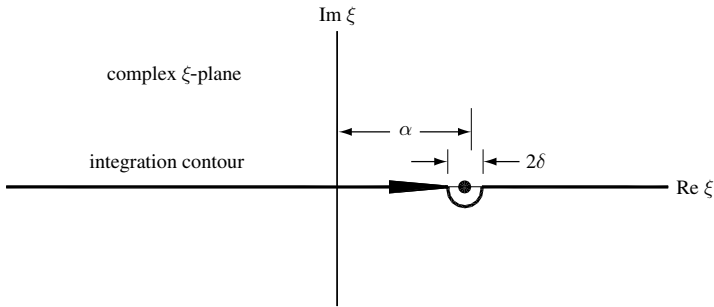


Fig. 5.6 Contour for evaluating plasma dispersion function.

The ξ integration contour can therefore be divided into three portions, namely (i) from $\xi = -\infty$ to $\xi = \alpha - \delta$, just to the left of the pole; (ii) a counterclockwise semi-circle of radius δ half-way around and *under* the pole (cf. Fig. 5.6); and (iii) a straight line from $\alpha + \delta$ to $+\infty$. The sum of the straight line segments (i) and (iii) in the limit $\delta \rightarrow 0$ is called the *principle part* of the integral and is denoted by a “P” in front of the integral sign. The semi-circle portion is *half* a residue and so makes a contribution that is just πi times the residue (rather than the standard $2\pi i$ for a complete residue). Hence, the plasma dispersion function for a pole slightly below the real axis is

$$Z(\alpha) = \frac{1}{\pi^{1/2}} \left[P \int_{-\infty}^{\infty} d\xi \frac{\exp(-\xi^2)}{(\xi - \alpha)} \right] + i\pi^{1/2} \exp(-\alpha^2), \quad (5.71)$$

where P means principle part of the integral. Equation (5.71) prescribes how to evaluate ill-defined integrals of the type we first noted in Eq. (5.32).

There are two important limiting situations for $Z(\alpha)$, namely $|\alpha| \gg 1$ (corresponding to the adiabatic fluid limit since $\omega/k \gg v_{T\sigma}$) and $|\alpha| \ll 1$ (corresponding to the isothermal fluid limit since $\omega/k \ll v_{T\sigma}$). Asymptotic evaluations of $Z(\alpha)$ are possible in both cases and are found as follows:

1. $\alpha \gg 1$ case.

Here, it is noted that the factor $\exp(-\xi^2)$ contributes significantly to the integral only when ξ is of order unity or smaller. In the important part of the integral where this exponential term is finite, $|\alpha| \gg \xi$. In this region of ξ the other factor in the integrand can be expanded as

$$\frac{1}{(\xi - \alpha)} = -\frac{1}{\alpha} \left(1 - \frac{\xi}{\alpha} \right)^{-1} = -\frac{1}{\alpha} \left[1 + \frac{\xi}{\alpha} + \left(\frac{\xi}{\alpha} \right)^2 + \left(\frac{\xi}{\alpha} \right)^3 + \left(\frac{\xi}{\alpha} \right)^4 + \dots \right]. \quad (5.72)$$

The expansion is carried to fourth order because of numerous cancelations that eliminate several of the lower order terms. Substitution of Eq. (5.72) into the integral in

Eq. (5.71) and noting that all odd terms in Eq. (5.72) do not contribute to the integral because the rest of the integrand is even gives

$$P \frac{1}{\pi^{1/2}} \int_{-\infty}^{\infty} d\xi \frac{\exp(-\xi^2)}{(\xi - \alpha)} = -\frac{1}{\alpha} \frac{1}{\pi^{1/2}} \int_{-\infty}^{\infty} d\xi \exp(-\xi^2) \times \left[1 + \left(\frac{\xi}{\alpha}\right)^2 + \left(\frac{\xi}{\alpha}\right)^4 + \dots \right]. \quad (5.73)$$

The “P” has been dropped from the right-hand side of Eq. (5.73) because there is no longer any problem with a singularity. These Gaussian-type integrals may be evaluated by taking successive derivatives with respect to a of the Gaussian

$$\frac{1}{\pi^{1/2}} \int d\xi \exp(-a\xi^2) = \frac{1}{a^{1/2}} \quad (5.74)$$

and then setting $a = 1$. Thus,

$$\frac{1}{\pi^{1/2}} \int d\xi \xi^2 \exp(-\xi^2) = \frac{1}{2}, \quad \frac{1}{\pi^{1/2}} \int d\xi \xi^4 \exp(-\xi^2) = \frac{3}{4} \quad (5.75)$$

so Eq. (5.73) becomes

$$P \frac{1}{\pi^{1/2}} \int_{-\infty}^{\infty} d\xi \frac{\exp(-\xi^2)}{(\xi - \alpha)} = -\frac{1}{\alpha} \left[1 + \frac{1}{2\alpha^2} + \frac{3}{4\alpha^4} + \dots \right]. \quad (5.76)$$

In summary, for $|\alpha| \gg 1$, the plasma dispersion function has the asymptotic form

$$Z(\alpha) = -\frac{1}{\alpha} \left[1 + \frac{1}{2\alpha^2} + \frac{3}{4\alpha^4} + \dots \right] + i\pi^{1/2} \exp(-\alpha^2). \quad (5.77)$$

2. $|\alpha| \ll 1$ case.

In order to evaluate the principle part integral in this regime, the variable $\eta = \xi - \alpha$ is introduced so that $d\eta = d\xi$. The integral may be evaluated as follows:

$$\begin{aligned} P \frac{1}{\pi^{1/2}} \int_{-\infty}^{\infty} d\xi \frac{\exp(-\xi^2)}{(\xi - \alpha)} &= \frac{1}{\pi^{1/2}} \int_{-\infty}^{\infty} d\eta \frac{e^{-\eta^2 - 2\alpha\eta - \alpha^2}}{\eta} \\ &= \frac{e^{-\alpha^2}}{\pi^{1/2}} \int_{-\infty}^{\infty} d\eta \frac{e^{-\eta^2}}{\eta} \\ &\quad \times \left[1 - 2\alpha\eta + \frac{(-2\alpha)^2}{2!} + \frac{(-2\alpha)^3}{3!} + \dots \right] \\ &= -2\alpha \frac{e^{-\alpha^2}}{\pi^{1/2}} \int_{-\infty}^{\infty} d\eta e^{-\eta^2} \left[1 + \frac{2\eta^2\alpha^2}{3} + \dots \right] \\ &= -2\alpha (1 - \alpha^2 + \dots) \left(1 + \frac{\alpha^2}{3} + \dots \right) \\ &= -2\alpha \left(1 - \frac{2\alpha^2}{3} + \dots \right), \end{aligned} \quad (5.78)$$

where in the third line all odd terms from the second line integrated to zero due to their symmetry. Thus, for $\alpha \ll 1$, the plasma dispersion function has the asymptotic limit

$$Z(\alpha) = -2\alpha \left(1 - \frac{2\alpha^2}{3} + \dots\right) + i\pi^{1/2} \exp(-\alpha^2). \quad (5.79)$$

5.3.6 Landau damping of electron plasma waves

The plasma susceptibilities given by Eq. (5.69) can now be evaluated. For $|\alpha| \gg 1$, using Eq. (5.77), and introducing the “frequency” $\omega = ip$ so that $\alpha = \omega/kv_{T\sigma}$ and $\alpha_i = \omega_i/kv_{T\sigma}$ the susceptibility is seen to be

$$\begin{aligned} \chi_\sigma &= \frac{1}{k^2 \lambda_{D\sigma}^2} \left\{ 1 + \alpha \left[-\frac{1}{\alpha} \left(1 + \frac{1}{2\alpha^2} + \frac{3}{4\alpha^4} + \dots \right) + i\pi^{1/2} \exp(-\alpha^2) \right] \right\} \\ &= \frac{1}{k^2 \lambda_{D\sigma}^2} \left\{ -\left(\frac{1}{2\alpha^2} + \frac{3}{4\alpha^4} + \dots \right) + i\alpha\pi^{1/2} \exp(-\alpha^2) \right\} \\ &= -\frac{\omega_{p\sigma}^2}{\omega^2} \left(1 + 3 \frac{k^2}{\omega^2} \frac{\kappa T_\sigma}{m_\sigma} + \dots \right) + i \frac{\omega}{kv_{T\sigma}} \frac{\pi^{1/2}}{k^2 \lambda_{D\sigma}^2} \exp(-\omega^2/k^2 v_{T\sigma}^2). \end{aligned} \quad (5.80)$$

Thus, if the root is such that $|\alpha| \gg 1$, the equation for the poles $D(p) = 1 + \chi_i + \chi_e = 0$ becomes

$$\begin{aligned} 1 - \frac{\omega_{pe}^2}{\omega^2} \left(1 + 3 \frac{k^2}{\omega^2} \frac{\kappa T_e}{m_e} + \dots \right) + i \frac{\omega}{kv_{Te}} \frac{\pi^{1/2}}{k^2 \lambda_{De}^2} \exp(-\omega^2/k^2 v_{Te}^2) \\ - \frac{\omega_{pi}^2}{\omega^2} \left(1 + 3 \frac{k^2}{\omega^2} \frac{\kappa T_i}{m_i} + \dots \right) + i \frac{\omega}{kv_{Ti}} \frac{\pi^{1/2}}{k^2 \lambda_{Di}^2} \exp(-\omega^2/k^2 v_{Ti}^2) = 0. \end{aligned} \quad (5.81)$$

This expression is similar to the previously obtained fluid dispersion relation, Eq. (4.32), but contains additional imaginary terms that did not exist in the fluid dispersion. Furthermore, Eq. (5.81) is not actually a dispersion relation. Instead, it is to be understood as the equation for the roots of $D(p)$. These roots determine the poles in $N(p)/D(p)$ producing the least damped oscillations resulting from some prescribed initial perturbation of the distribution function. Since $\omega_{pe}^2/\omega_{pi}^2 = m_i/m_e$, and in general $v_{Ti} \ll v_{Te}$, both the real and imaginary parts of the ion terms are much smaller than the corresponding electron terms. On dropping the ion terms, the expression becomes

$$1 - \frac{\omega_{pe}^2}{\omega^2} \left(1 + 3 \frac{k^2}{\omega^2} \frac{\kappa T_e}{m_e} + \dots \right) + i \frac{\omega}{kv_{Te}} \frac{\pi^{1/2}}{k^2 \lambda_{De}^2} \exp(-\omega^2/k^2 v_{Te}^2) = 0. \quad (5.82)$$

Recalling that $\omega = ip$ is complex, we write $\omega = \omega_r + i\omega_i$ and then proceed to find the complex ω that is the root of Eq. (5.82). Although it would not be particularly

difficult to substitute $\omega = \omega_r + i\omega_i$ into Eq. (5.82) and then manipulate the coupled real and imaginary parts of this equation to solve for ω_r and ω_i , it is better to take this analysis as an opportunity to introduce a more general way for solving equations of this sort.

Equation (5.82) can be written as

$$D(\omega_r + i\omega_i) = D_r(\omega_r + i\omega_i) + iD_i(\omega_r + i\omega_i) = 0, \quad (5.83)$$

where D_r is the part of D that does not explicitly contain i and D_i is the part that does explicitly contain i . Thus,

$$D_r = 1 - \frac{\omega_{pe}^2}{\omega^2} \left(1 + 3 \frac{k^2}{\omega^2} \frac{\kappa T_e}{m_e} + \dots \right), \quad D_i = \frac{\omega}{kv_{Te}} \frac{\pi^{1/2}}{k^2 \lambda_{De}^2} \exp(-\omega^2/k^2 v_{Te}^2). \quad (5.84)$$

Since the oscillation has been assumed to be weakly damped, $\omega_i \ll \omega_r$ and so Eq. (5.83) can be Taylor expanded in the small quantity ω_i ,

$$D_r(\omega_r) + i\omega_i \left(\frac{dD_r}{d\omega} \right)_{\omega=\omega_r} + i \left[D_i(\omega_r) + i\omega_i \left(\frac{dD_i}{d\omega} \right)_{\omega=\omega_r} \right] = 0. \quad (5.85)$$

Since $\omega_i \ll \omega_r$, the real part of Eq. (5.85) is

$$D_r(\omega_r) \simeq 0. \quad (5.86)$$

Balancing the two imaginary terms in Eq. (5.85) gives

$$\omega_i = - \frac{D_i(\omega_r)}{\frac{dD_r}{d\omega}}. \quad (5.87)$$

Thus, Eqs. (5.86) and (5.84) give the real part of the frequency as

$$\omega_r^2 = \omega_{pe}^2 \left(1 + 3 \frac{k^2}{\omega_r^2} \frac{\kappa T_e}{m_e} \right) \simeq \omega_{pe}^2 (1 + 3k^2 \lambda_{De}^2), \quad (5.88)$$

while Eqs. (5.87) and (5.84) give the imaginary part of the frequency, called the *Landau damping*, as

$$\begin{aligned} \omega_i &= - \sqrt{\frac{\pi}{8}} \frac{\omega_{pe}}{k^3 \lambda_{De}^3} \exp(-\omega^2/k^2 v_{Te}^2) \\ &= - \sqrt{\frac{\pi}{8}} \frac{\omega_{pe}}{k^3 \lambda_{De}^3} \exp[-(1 + 3k^2 \lambda_{De}^2)/2k^2 \lambda_{De}^2]. \end{aligned} \quad (5.89)$$

Since the least damped oscillation goes as $\exp(pt) = \exp(-i\omega t) = \exp(-i(\omega_r + i\omega_i)t) = \exp(-i\omega_r t + \omega_i t)$ and Eq. (5.89) gives a negative ω_i , this is indeed a damping. It is interesting to note that while Landau damping was proposed

theoretically by Landau in 1949, it took sixteen years before Landau damping was verified experimentally (Malmberg and Wharton 1964).

What is meant by weak damping vs. strong damping? In order to calculate ω_i it was assumed that ω_i is small compared to ω_r suggesting perhaps that ω_i is unimportant. However, even though small, ω_i can be important, because the factor 2π affects the real and imaginary parts of the wave phase differently. Suppose for example that the imaginary part of the frequency is $1/2\pi \sim 1/6$ the magnitude of the real part. This ratio is surely small enough to justify the Taylor expansion used in Eq. (5.85) and also to justify the assumption that the pole p_j corresponding to this mode is only slightly to the left of the imaginary p axis. Let us calculate how much the wave is attenuated in one period $\tau = 2\pi/\omega_r$. This attenuation will be $\exp(-|\omega_i|\tau) = \exp(-2\pi/6) \sim \exp(-1) \sim 0.3$. Thus, the wave amplitude decays to one third of its original value in just one period, which is certainly important.

5.3.7 Power relationships

It is premature to calculate the power associated with wave damping, because we do not yet know how to add up all the energy in the wave. Nevertheless, if we are willing to assume temporarily that the wave energy is entirely in the wave electric field (it turns out there is also energy in coherent particle motion – to be discussed in Chapter 14), it is seen that the power being lost from the wave electric field is

$$\begin{aligned} P_{\text{wavelost}} &\sim \frac{d}{dt} \left\langle \frac{\epsilon_0 E_{\text{wave}}^2}{2} \right\rangle \sim \frac{d}{dt} \left[\frac{\epsilon_0 |E_{\text{wave}}|^2}{4} \exp(-2|\omega_i|t) \right] = -\frac{|\omega_i| \epsilon_0 E_{\text{wave}}^2}{2} \\ &= \sqrt{\frac{\pi}{8}} \frac{\omega_{pe}}{2k^3 \lambda_{De}^3} \exp(-\omega^2/k^2 v_{T\sigma}^2) \epsilon_0 E_{\text{wave}}^2, \quad (5.90) \end{aligned}$$

where $\langle E_{\text{wave}}^2 \rangle = |E_{\text{wave}}|^2 \langle \cos(kx - \omega t) \rangle = |E_{\text{wave}}|^2/2$ has been used. However, in Section 3.8, it was shown that the energy gained by untrapped resonant particles in a wave is

$$\begin{aligned} P_{\text{partgain}} &= \frac{-\pi m \omega}{2k^2} \left(\frac{q E_{\text{wave}}}{m} \right)^2 \left[\frac{d}{dv_0} f(v_0) \right]_{v_0=\omega/k} \\ &= \frac{-\pi m \omega}{2k^2} \left(\frac{q E_{\text{wave}}}{m} \right)^2 \left[\frac{d}{dv_0} \left\{ \left(\frac{m}{2\pi \kappa T} \right)^{1/2} n_0 \exp\left(-\frac{mv^2}{2\kappa T}\right) \right\} \right]_{v_0=\omega/k} \\ &= \frac{\pi m \omega}{2k^2} \left(\frac{q E_{\text{wave}}}{m} \right)^2 \left(\frac{m}{2\pi \kappa T} \right)^{1/2} \left(\frac{m \omega}{\kappa T k} \right) n_0 \exp\left(-\frac{\omega^2}{k^2 v_{T\sigma}^2}\right); \quad (5.91) \end{aligned}$$

using $\omega \sim \omega_{pe}$ this is seen to be the same as Eq. (5.90) except for a factor of two. We shall see later that this factor of two comes from the fact that the wave

electric field actually contains *half* the energy of the electron plasma wave, with the other half in coherent particle motion, so the true power loss rate is really twice that given in Eq. (5.90).

5.3.8 Landau damping for ion acoustic waves

Ion acoustic waves resulted from a two-fluid analysis in the regime where the wave-phase velocity was intermediate between the electron and ion thermal velocities. In this situation the electrons behave isothermally and the ions behave adiabatically. This suggests there might be another root of $D(p)$ if $|\alpha_e| \ll 1$ and $|\alpha_i| \gg 1$ or equivalently $v_{Ti} \ll \omega/k \ll v_{Te}$. From Eqs. (5.69) and (5.79), the susceptibility for $|\alpha| \ll 1$ is found to be

$$\begin{aligned}\chi_\sigma &= \frac{1}{k^2 \lambda_{D\sigma}^2} [1 + \alpha Z(\alpha)] \\ &= \frac{1}{k^2 \lambda_{D\sigma}^2} \left\{ 1 - 2\alpha^2 \left(1 - \frac{2\alpha^2}{3} + \dots \right) + i\alpha \pi^{1/2} \exp(-\alpha^2) \right\} \\ &\simeq \frac{1}{k^2 \lambda_{D\sigma}^2} + i \frac{\alpha}{k^2 \lambda_{D\sigma}^2} \pi^{1/2} \exp(-\alpha^2).\end{aligned}\quad (5.92)$$

Using Eq. (5.92) for the electron susceptibility and Eq. (5.80) for the ion susceptibility gives

$$\begin{aligned}D(\omega) &= 1 + \frac{1}{k^2 \lambda_{De}^2} + i \frac{\omega}{k v_{Te}} \frac{\pi^{1/2}}{k^2 \lambda_{De}^2} \exp(-\omega^2/k^2 v_{Te}^2) \\ &\quad - \frac{\omega_{pi}^2}{\omega^2} \left(1 + 3 \frac{k^2}{\omega^2} \frac{\kappa T_i}{m_i} + \dots \right) + i \frac{\omega}{k v_{Ti}} \frac{\pi^{1/2}}{k^2 \lambda_{Di}^2} \exp(-\omega^2/k^2 v_{Ti}^2).\end{aligned}\quad (5.93)$$

On applying the Taylor expansion technique discussed in conjunction with Eqs. (5.86) and (5.87) we find that ω_r is the root of

$$D_r(\omega_r) = 1 + \frac{1}{k^2 \lambda_{De}^2} - \frac{\omega_{pi}^2}{\omega_r^2} \left(1 + 3 \frac{k^2}{\omega_r^2} \frac{\kappa T_i}{m_i} \right) = 0, \quad (5.94)$$

i.e.,

$$\omega_r^2 = \frac{k^2 c_s^2}{1 + k^2 \lambda_{De}^2} \left(1 + 3 \frac{k^2}{\omega_r^2} \frac{\kappa T_i}{m_i} \right) \simeq \frac{k^2 c_s^2}{1 + k^2 \lambda_{De}^2} + 3k^2 \frac{\kappa T_i}{m_i}. \quad (5.95)$$

Here, as in the two-fluid analysis of ion acoustic waves, $c_s^2 = \omega_{pi}^2 \lambda_{De}^2 = \kappa T_e / m_i$ has been defined. The imaginary part of the frequency is found to be

$$\begin{aligned}
 \omega_i &\simeq -\frac{D_i(\omega_r)}{dD_r/d\omega} \\
 &= -\frac{\omega \pi^{1/2}}{k^3} \left[\frac{\frac{1}{\lambda_{De}^2 v_{Te}} \exp(-\omega^2/k^2 v_{Te}^2) + \frac{1}{\lambda_{Di}^2 v_{Ti}} \exp(-\omega^2/k^2 v_{Ti}^2)}{2\omega_{pi}^2/\omega^3} \right] \\
 &= -\frac{\omega^4}{k^3 c_s^3} \sqrt{\frac{\pi}{8}} \left[\sqrt{\frac{m_e}{m_i}} + \left(\frac{T_e}{T_i}\right)^{3/2} \exp(-\omega^2/k^2 v_{Ti}^2) \right] \\
 &= -\frac{|\omega_r|}{(1+k^2 \lambda_{De}^2)^{3/2}} \sqrt{\frac{\pi}{8}} \left[\sqrt{\frac{m_e}{m_i}} + \left(\frac{T_e}{T_i}\right)^{3/2} \exp\left(-\frac{T_e/2T_i}{1+k^2 \lambda_{De}^2} - \frac{3}{2}\right) \right]. \quad (5.96)
 \end{aligned}$$

The dominant Landau damping comes from the ions, since the electron Landau damping term has the small factor $\sqrt{m_e/m_i}$. If $T_e \gg T_i$ the ion term also becomes small because $x^{3/2} \exp(-x) \rightarrow 0$ as x becomes large. Hence, strong ion Landau damping occurs when T_i approaches T_e and so ion acoustic waves can only propagate without extreme attenuation if the plasma has $T_e \gg T_i$. Landau damping of ion acoustic waves was first observed experimentally by Wong, Motley and D'Angelo (1964).

5.3.9 The Plemelj formula

The Landau method showed that the correct way to analyze problems that lead to ill-defined integrals such as Eq. (5.35) is to pose the problem as an initial value problem rather than as a steady-state situation. The essential result of the Landau method can be summarized by the Plemelj formula

$$\lim_{\varepsilon \rightarrow 0} \frac{1}{\xi - a \mp i|\varepsilon|} = \mathcal{P} \frac{1}{\xi - a} \pm i\pi \delta(\xi - a), \quad (5.97)$$

which is a prescription showing how to deal with singular integrands of the form appearing in the plasma dispersion function. From now on, instead of repeating the lengthy Laplace transform analysis, we instead will use the less cumbersome, but formally incorrect, Fourier method and then invoke Eq. (5.97) as a “patch” to resolve any ambiguities regarding integration contours.

5.4 The Penrose criterion

The analysis so far showed that electrostatic plasma waves are subject to Landau damping, a collisionless attenuation proportional to $[\partial f / \partial v]_{v=\omega/k}$, and that this damping is consistent with the calculation of power input to particles by an electrostatic wave. Since a Maxwellian distribution function has a negative slope, its associated Landau damping is always a true wave damping. This is consistent with the physical picture developed in the single particle analysis, which showed that energy is transferred from wave to particles if there are more slow than fast particles in the vicinity of the wave phase velocity. What happens if there is a non-Maxwellian distribution function, in particular one where there are more fast particles than slow particles in the vicinity of the wave-phase velocity, i.e., $[\partial f / \partial v]_{v=\omega/k} > 0$? Because $f(v) \rightarrow 0$ as $|v| \rightarrow \infty$, f can only have a positive slope for a finite range of positive velocity and can only have a negative slope for a finite range of negative velocity; in particular, positive slopes of the distribution function must always be located to the left of a localized maximum in $f(v)$ in the $v > 0$ velocity-space region. A localized maximum in $f(v)$ corresponds to a beam of fast particles superimposed on a (possible) background of particles having a monotonically decreasing $f(v)$. Can the Landau damping process be run in reverse and so provide Landau growth, i.e., wave instability? The answer is yes. We will now discuss a criterion due to Penrose (1960) that shows how strong a beam must be to give Landau instability.

The procedure used to derive Eq. (5.32) is repeated, giving

$$1 + \frac{q^2}{k^2 m \epsilon_0} \int_{-\infty}^{\infty} \frac{k}{(\omega - kv)} \frac{\partial f_0}{\partial v} dv = 0, \quad (5.98)$$

which may be recast as

$$k^2 = Q(z), \quad (5.99)$$

where

$$Q(z) = \frac{q^2}{m \epsilon_0} \int_{-\infty}^{\infty} \frac{1}{(v - z)} \frac{\partial f_0}{\partial v} dv \quad (5.100)$$

is a complex function of the complex variable $z = \omega/k$. The wavenumber k is assumed to be a positive real quantity and the Plemelj formula will be used to resolve the ambiguity due to the singularity in the integrand.

The left-hand side of Eq. (5.99) is, by assumption, always real and positive for any choice of k . A solution of this equation can therefore always be found if $Q(z)$ is simultaneously pure real and positive. The actual magnitude of $Q(z)$ does not matter, since the magnitude of k^2 can be adjusted to match the magnitude of $Q(z)$.

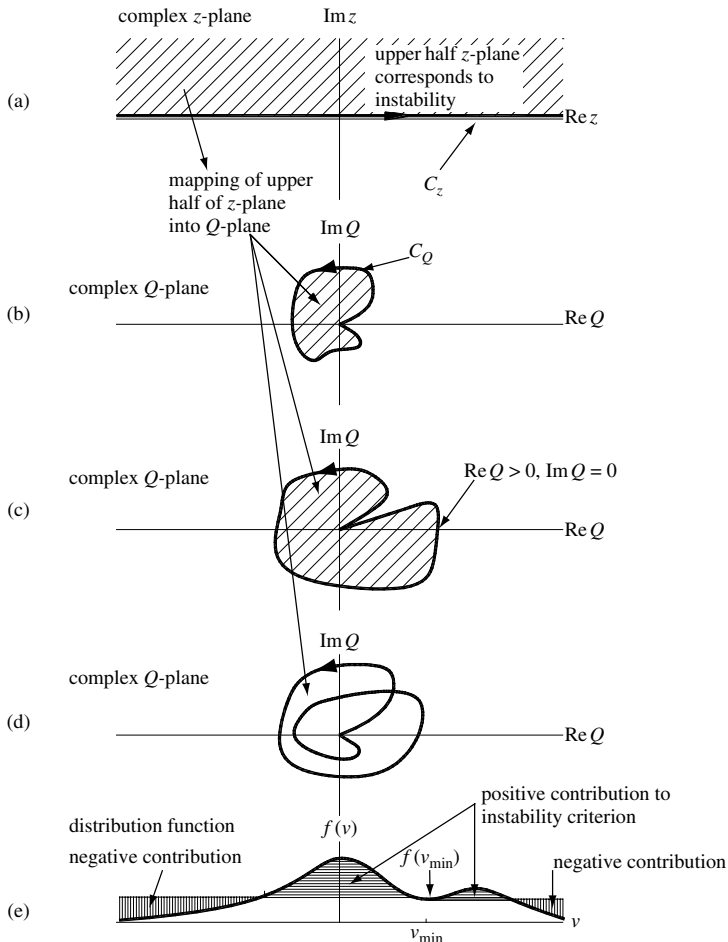


Fig. 5.7 Penrose criterion: (a)–(d) mappings, (e) instability criterion.

The function $Q(z)$ may be interpreted as a mapping from the complex z -plane to the complex Q -plane. Because solutions of Eq. (5.99) giving instability are those for which $\text{Im } \omega > 0$, the upper half of the complex z -plane corresponds to instability and the real z axis represents the dividing line between stability and instability. Let us consider a straight-line contour C_z parallel to the real z axis, and slightly above. As shown in Fig. 5.7(a) this contour can be prescribed as $z = z_r + i\delta$, where δ is a small constant and z_r ranges from $-\infty$ to $+\infty$.

The function $Q(z) \rightarrow 0$ when $z \rightarrow \pm\infty$ and so, as z is moved along the C_z contour, the corresponding path C_Q traced in the Q -plane must start at the origin and end at the origin. Furthermore, since Q can be evaluated using the Plemelj

formula, it is seen that Q is finite for all z on the path C_z . Thus, C_Q is a continuous finite curve starting at the Q -plane origin and ending at this same origin as shown by the various possible mappings sketched in Figs. 5.7(b), (c), and (d).

The upper half z -plane maps to the area inside the curve C_Q . If C_Q is of the form shown in Fig. 5.7(b), then $Q(z)$ never takes on a positive real value for z being in the upper half z -plane; thus a curve of this form cannot give a solution to Eq. (5.99) corresponding to an instability. However, curves of the form sketched in Figs. 5.7 (c) and (d) do have $Q(z)$ taking on positive real values and so do correspond to unstable solutions. Marginally unstable situations correspond to where C_Q crosses the positive real Q axis, since C_Q is a mapping of C_z which was the set of marginally unstable frequencies.

Let us therefore focus attention on what happens when C_Q crosses the positive real Q axis. Using the Plemelj formula on Eq. (5.100) it is seen that

$$\text{Im } Q = \frac{q^2}{m\epsilon_0} \pi \left[\frac{\partial f_0}{\partial v} \right]_{v=\omega_r/k} \quad (5.101)$$

and, on moving along C_Q from a point just below the real Q axis to just above the real Q axis, $\text{Im } Q$ goes from being negative to positive. Thus, $[\partial f_0 / \partial v]_{v=\omega_r/k}$ changes from being negative to positive, so that on the positive real Q axis f_0 is a *minimum* at some value $v = v_{\min}$ (here the subscript “min” means the value of v for which f_0 is at a minimum and not v itself is at a minimum). A Taylor expansion about this minimum gives

$$f(v) = f[v_{\min} + (v - v_{\min})] = f(v_{\min}) + 0 + \frac{(v - v_{\min})^2}{2} f''(v_{\min}) + \dots \quad (5.102)$$

Since $f(v_{\min})$ is a constant, it is permissible to write

$$\frac{\partial f_0}{\partial v} = \frac{\partial}{\partial v} [f_0(v) - f_0(v_{\min})]. \quad (5.103)$$

This innocuous insertion of $f(v_{\min})$ makes it easy to integrate Eq. (5.100) by parts

$$\begin{aligned} Q(z = v_{\min}) &= \frac{q^2}{m\epsilon_0} \left\{ \text{P} \int_{-\infty}^{\infty} dv \frac{\frac{\partial}{\partial v} [f(v) - f(v_{\min})]}{(v - v_{\min})} + i\pi \left[\frac{\partial}{\partial v} f(v) \right]_{v=v_{\min}} \right\} \\ &= \frac{q^2}{m\epsilon_0} \text{P} \int_{-\infty}^{\infty} dv \frac{1}{(v - v_{\min})^2} [f(v) - f(v_{\min})] \\ &= \frac{q^2}{m\epsilon_0} \int_{-\infty}^{\infty} dv \frac{1}{(v - v_{\min})^2} \left[0 + \frac{(v - v_{\min})^2}{2} f''(v_{\min}) + \dots \right]; \end{aligned} \quad (5.104)$$

in the second line, advantage has been taken of the imaginary part being zero by assumption, and in the third line the “P” for principle part has been dropped because there is no longer a singularity at $v = v_{\min}$. In fact, since the leading term of $f(v) - f(v_{\min})$ is proportional to $(v - v_{\min})^2$, this qualifying “P” can also be dropped from the second line. The requirement for marginal instability can be summarized as: $f(v)$ has a minimum at $v = v_{\min}$, and the value of Q is positive, i.e.,

$$Q(v_{\min}) = \frac{q^2}{m\epsilon_0} \int_{-\infty}^{\infty} dv \left[\frac{f(v) - f(v_{\min})}{(v - v_{\min})^2} \right] > 0. \quad (5.105)$$

This is just a weighted measure of the strength of the bump in f located to the right of the minimum as shown in Fig. 5.7(e). The hatched areas with horizontal lines make positive contributions to Q , while the hatched areas with vertical lines make negative contributions. These contributions are weighted according to how far they are from $v_{\min}$ by the factor $(v - v_{\min})^{-2}$.

The Penrose criterion extends the two-stream instability analysis to an arbitrary distribution function containing finite temperature beams.

5.5 Assignments

1. Show that the electrostatic dispersion relation for electrons streaming with velocity $\mathbf{v}_0$ through stationary ions is

$$1 - \frac{\omega_{pi}^2}{\omega^2} - \frac{\omega_{pe}^2}{(\omega - \mathbf{k} \cdot \mathbf{v}_0)^2} = 0.$$

- (a) Show that instability begins when

$$\left(\frac{\mathbf{k} \cdot \mathbf{v}_0}{\omega_{pe}} \right)^2 < \left[1 + \left(\frac{m_e}{m_i} \right)^{1/3} \right]^3.$$

- (b) Split the frequency into its real and imaginary parts so that $\omega = \omega_r + i\omega_i$. Show that the instability has maximum growth rate

$$\frac{\omega_i}{\omega_{pe}} = \frac{\sqrt{3}}{2} \left(\frac{m_e}{2m_i} \right)^{1/3}.$$

What is the value of kv_0/ω_{pe} when the instability has maximum growth rate? Sketch the dependence of ω_i/ω_{pe} on kv_0/ω_{pe} . (Hint: define non-dimensional variables $\epsilon = m_e/m_i$, $z = \omega/\omega_{pe}$, and $\lambda = kv_0/\omega_{pe}$. Let $z = x + iy$ and look for the maximum y satisfying the dispersion. A particularly neat way to solve the dispersion is to solve the dispersion for the imaginary part of λ , which of course is zero, since by assumption k is real. Take advantage of the assumption $\epsilon \ll 1$ to find a relatively simple expression involving y . Maximize y with respect to x and then find the respective values of x , y , and λ at this point of maximum y .)

2. Prove the Plemelj formula.
3. Suppose

$$\mathbf{E}(\mathbf{x}, t) = \int \tilde{\mathbf{E}}(\mathbf{k}) e^{i\mathbf{k}\cdot\mathbf{x} - i\omega(\mathbf{k})t} d\mathbf{k}, \quad (5.106)$$

where

$$\omega = \omega_r(\mathbf{k}) + i\omega_i(\mathbf{k})$$

is determined by an appropriate dispersion relation. Assuming $\mathbf{E}(\mathbf{x}, t)$ is a real quantity, show, by comparing Eq. (5.106) to its complex conjugate, that $\omega_r(\mathbf{k})$ must always be an odd function of $\mathbf{k}$, while ω_i must always be an even function of $\mathbf{k}$. (Hint: note that the left-hand side of Eq. (5.106) is real by assumption, and so the right-hand side must also be real. Take the complex conjugate of both sides and replace the dummy variable of integration $\mathbf{k}$ by $-\mathbf{k}$ so that $d\mathbf{k} \rightarrow -d\mathbf{k}$ and the $\pm\infty$ limits of integration are also interchanged.)

4. Plot the real and imaginary parts of the plasma dispersion function. Plot the real and imaginary parts of the susceptibilities.
5. Is it possible to have a propagating electrostatic plasma wave that has $k\lambda_{De} \gg 1$. Hint: consider Landau damping.
6. Plot the potential versus time in units of the real period of an electron plasma wave for various values of $\omega/k\sqrt{\kappa T_e/m_e}$ showing the onset of Landau damping.
7. Plot ω_i/ω_r for ion acoustic waves for various values of T_e/T_i and show that these waves have strong ion Landau damping when the ion temperature approaches the electron temperature.
8. Landau instability for ion acoustic waves. Plasmas with $T_e > T_i$ support propagation of ion acoustic waves; these waves are Landau damped by both electrons and ions. However, if there is a sufficiently strong current J flowing in the plasma giving a relative streaming velocity $u_0 = J/ne$ between the ions and electrons, the Landau damping can operate in reverse, and give a Landau growth. This can be seen by moving to the ion frame, in which case the electrons appear as an offset Gaussian. If the offset is large enough it will be possible to have $[\partial f_e/\partial v]_{v \sim c_s} > 0$, giving more fast than slow particles at the wave-phase velocity. Now, since f_e is a Gaussian with its center shifted to be at u_0 , show that if $u_0 > \omega/k$ the portion of f_e immediately to the left of u_0 will have positive slope and so lead to instability. These qualitative ideas can easily be made quantitative, by considering a 1-D equilibrium where the ion distribution is

$$f_{i0} = \frac{n_0}{\pi^{1/2} v_{Ti}} e^{-v^2/v_{Ti}^2}$$

and the drifting electron distribution is

$$f_{e0} = \frac{n_0}{\pi^{1/2} v_{Te}} e^{-(v-u_0)^2/v_{Te}^2}.$$

The ion susceptibility will be the same as before, but to determine the electron susceptibility we must reconsider the linearized Vlasov equation

$$\frac{\partial f_{e1}}{\partial t} + v \frac{\partial f_{e1}}{\partial x} - \frac{q_e}{m_e} \frac{\partial \phi_1}{\partial x} \frac{\partial}{\partial v} \left[\frac{n_0}{\pi^{1/2} v_{Te}} e^{-(v-u_0)^2/v_{Te}^2} \right] = 0.$$

This equation can be simplified by defining $v' = v - u_0$. Show that the electron susceptibility becomes

$$\chi_e = \frac{1}{k^2 \lambda_{De}^2} [1 + \alpha Z(\alpha)],$$

where now $\alpha = (\omega - k u_0)/k v_{Te}$. Suppose $T_e \gg T_i$ so that the electron Landau damping term dominates. Show that if $u_0 > \omega_r/k$ the electron imaginary term will reverse sign and give instability.

9. Suppose a current I flows in a long cylindrical plasma of radius a , density n , ion mass m_i for which $T_e \gg T_i$. Write a criterion for ion acoustic instability in terms of an appropriate subset of these parameters. Suppose a cylindrical mercury plasma with $T_e = 1 - 2$ eV, $T_i = 0.1$ eV, diameter 2.5 cm, carries a current of $I = 0.35$ amps. At what density would an ion acoustic instability be expected to develop? Does this configuration remind you of an everyday object? Hint: there are several hanging from the ceiling of virtually every classroom.